

Analysis Report — balearn_multi_Wbd5

Mode: audit_report

File: balearn_multi_Wbd5
Rows: 4 | Fields: 5

Key Metrics

Metric	Value
Total Revenue	645000

Dataset Overview

- File: balearn_multi_Wbd5
- Records: 4 | Fields: 5
- Missing values: None
- Duplicate rows: 0

Fields

store_id (object), revenue (int64), cost (int64), region (object), size (object)

Field Semantics

Field	Inferred Meaning	Confidence
store_id	Unique identifier	high
revenue	Revenue / total sales	high
cost	Marketing spend / cost	high
region	Geographic region	high
size	Store size / square footage	high

Key Findings

revenue Top/Bottom by region

Evidence: Top: West (220,000.00), Bottom: North (95,000.00). Range ratio: 2.3x

Interpretation: region is a strong differentiator for revenue. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top region segments and apply best practices to lower performers.

cost Top/Bottom by region

Evidence: Top: West (165,000.00), Bottom: North (68,000.00). Range ratio: 2.4x

Interpretation: region is a strong differentiator for cost. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top region segments and apply best practices to lower performers.

revenue Top/Bottom by size

Evidence: Top: Large (370,000.00), Bottom: Small (95,000.00). Range ratio: 3.9x

Interpretation: size is a strong differentiator for revenue. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top size segments and apply best practices to lower performers.

cost Top/Bottom by size

Evidence: Top: Large (270,000.00), Bottom: Small (68,000.00). Range ratio: 4.0x

Interpretation: size is a strong differentiator for cost. The gap between top and bottom segments suggests significant optimization opportunity.

Action: Investigate what drives success in top size segments and apply best practices to lower performers.

Thinking Models

1. 分解思维 (Decomposition)

Evidence: 将数据按照 Geographic region, Store size / square footage 等维度拆解分析。各分组间的 Revenue / total sales, Marketing spend / cost 存在显著差异，整体指标无法反映细分表现。

Takeaway: 永远先拆解再下结论。汇总指标会隐藏关键的分组差异。

2. 分层差异 (Segment Divergence)

Evidence: 数据集包含 2 个分类维度 (Geographic region, Store size / square footage)。各维度下的分组均值差异倍数可达数倍甚至数十倍。

Takeaway: 不要相信平均值。检查分组间的差异倍数，差异越大，平均值越具有误导性。

3. 间接推断 (Proxy Inference)

Evidence: 2 个数值字段 (含 Revenue / total sales, Marketing spend / cost) 中，部分字段可作为业务健康度的代理指标。没有直接的数据时，用可观察的指标间接推断。

Takeaway: 选择合适的代理变量，比等待完美数据更重要。

4. 杠杆点识别 (Leverage Point)

Evidence: 通过按 Geographic region, Store size / square footage 分组排名分析，识别出高贡献/高差异的分组。这些分组同时具备"影响面广"和"改善空间大"两个特点。

Takeaway: 聚焦高份额 + 高差异的分组，资源投入产出比最高。

5. 约束 vs 偏好 (Constraint vs Preference)

Evidence: 分组间的差异可能是结构性约束（用户无法选择）或偏好（用户主动选择）导致。需要结合业务背景判断。

Takeaway: 约束型问题需要改变产品/流程；偏好型问题需要改变营销/定位。诊断清楚再行动。

Learning Guide

Concepts Learned

- 业务指标的定义和计算口径
- 数据质量对分析结论的影响

Methods Learned

- 分组聚合分析
- 趋势分析
- 对比分析 (Top/Bottom)

Metrics Learned

- Total Revenue

Reusable Thinking Patterns

- 分解思维
- 分层差异
- 间接推断
- 杠杆点识别
- 约束 vs 偏好

Self-Check Questions

Q1: What total metric can be decomposed, and by which dimensions?

Q2: After decomposition, how large are the subgroup differences?

Q3: What missing key information can be inferred by a proxy variable?

Q4: Is the observed difference a preference or a constraint?

Q5: Which segment is both high-share and high-difference (a leverage point)?

Metrics Glossary

Metric	Value	Formula
Total Revenue	645000	SUM(revenue) = 645,000.00

Total Revenue

Meaning: Total sales value

Business Use: Core measure of business scale

Data Tables

数据表关系

表名	列名	列名	列名
sales	4	3	store_id
stores	4	3	store_id

按region看revenue

region	revenue_表名	revenue_表名	表名
West	220000	220000.0	1
South	180000	180000.0	1
East	150000	150000.0	1
North	95000	95000.0	1

按region看cost

region	cost_表名	cost_表名	表名
West	165000	165000.0	1
South	129000	129000.0	1
East	105000	105000.0	1
North	68000	68000.0	1

按size看revenue

size	revenue_表名	revenue_表名	表名
Large	370000	185000.0	2
Medium	180000	180000.0	1
Small	95000	95000.0	1

按size看cost

size	cost_表名	cost_表名	表名
Large	270000	135000.0	2
Medium	129000	129000.0	1
Small	68000	68000.0	1

revenue分布

统计量	revenue
count	4.0
mean	161250.0
std	52658.49
min	95000.0
25%	136250.0
50%	165000.0
75%	190000.0
max	220000.0

cost分布

■■■	cost
count	4.0
mean	116750.0
std	40795.22
min	68000.0
25%	95750.0
50%	117000.0
75%	138000.0
max	165000.0

关联键检测

■A	■B	■■■	■■■■	■■■(A/B)	■■
sales	stores	store_id	one_to_one	100.0% / 100.0%	safe

关联计划

■■	■A	■B	■	■■■■	■■
Step 1	sales	stores	store_id	left	safe

Recommendations

优先关注 Top-Bottom 差距大的维度

Action: 对排名靠前和靠后的分组分别分析，找出最佳实践和问题根因。将成功经验横向复制。

Metric to Track: 分组间的差距倍数变化

Data Needed: 细分到分组的详细数据

建立定期监控机制

Action: 对核心数值字段建立定期（周/月）监控，及时发现异常波动。

Metric to Track: 核心指标的环比和同比增长率

Data Needed: 连续时间序列数据